



OPEN Encapsulation of *Beauveria bassiana* conidia as a new strategy for the biological control of *Aedes aegypti* larvae

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The virulence of encapsulated fungal conidia against *Aedes aegypti* larvae was investigated. Molecular studies confirmed that the fungal isolate used here was *Beauveria bassiana*. Different conidial concentrations were tested. A concentration of 1×10^8 conidia mL^{-1} was the most effective, resulting in 7% larval survival after 7 days. Next, alginate capsules (0.65%) containing conidia were prepared with different densities of calcium chloride (0.01 M, 0.009 M, and 0.008 M CaCl_2) and tested against larvae. Furthermore, groups of capsules were prepared with bird diet to act as an attractant. All capsule densities tested reduced larval survival (ranging from 22 to 67%). However, capsules with 0.008 M CaCl_2 were the most effective. Furthermore, fungus-only capsules were more efficient when compared to those containing bird diet. Laboratory and semi-field bioassays were conducted using mixtures of capsules with different densities. In the laboratory, survival ranged from 26 to 53%, whereas in semi-field conditions, 35%, and 80% survival was observed for groups exposed to fungus-only capsules or capsules containing diet, respectively. Histopathological studies of larvae exposed to capsules showed the presence of the fungus in the digestive tract and visible damage to enterocytes. These findings offer new insights into the biological control of *Ae. aegypti* larvae.

Keywords capsules, alginate, dengue, biocontrol, entomopathogenic fungi, enterocytes

Entomopathogenic fungi (EPF) are biological control agents that can be deployed against mosquitoes of public health concern¹. These fungi could be incorporated into integrated vector management programs. Integrating EPF into comprehensive vector control strategies holds particular potential, with a focus on addressing the constant menace posed by *Aedes aegypti* (Diptera: Culicidae). This species is responsible for transmitting the arthropod-borne viruses dengue, Zika, and chikungunya, affecting millions of people worldwide².

Currently, a vaccine against dengue is available³; however, it is expensive and not accessible for use globally⁴. Thus, integrated vector control should be considered as a primary intervention against mosquitoes. The most commonly used approach for controlling *Ae. aegypti* involves environmental management to eliminate breeding sites, such as tires and water containers; and the application of synthetic chemical insecticides. However, resistance to various molecules has been documented for *Ae. aegypti* worldwide^{5–9}. Moreover, synthetic insecticides may have adverse effects on non-target species, including humans¹⁰. Therefore, additional tools are needed to control mosquito populations and reduce the transmission of arboviruses.

Entomopathogenic fungi have the potential to control *Ae. aegypti* eggs¹¹, larvae¹², pupae¹³, and adults¹⁴. EPF are widespread in nature and exhibit high levels of genetic diversity, allowing the selection of highly virulent and selective strains to control specific insect species^{15,16}. Currently, EPF are employed commercially in agriculture¹⁷. However, no products are available to control insects of public health concern.

Beauveria (Balsamo) Vuillemin (Hypocreales: Cordyciptaceae) is one of the most studied EPF genera for use against mosquitoes^{18,19}. Unlike other entomopathogenic microorganisms, EPF can infect their host by

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penetrating the cuticle^{20,21}. After conidia adhesion, the fungus produces a germ tube and a penetration peg to overcome the physical barrier of the integument. Once inside the hemocoel, the fungus differentiates into hyphal bodies/blastospores simultaneously producing an array of toxic secondary metabolites, downregulating the immunological response of the target insect²² and blocking calcium channels, leading to flaccid paralysis²³. Subsequently, the fungus colonizes the insect tissues, absorbing nutrients, and finally reemerges from the insect cadaver to sporulate¹⁵.

Some studies involving fungal encapsulation have been shown to improve the efficiency of EPF conidia for use in biocontrol strategies, increasing viability, virulence, and shelf life^{24,25}. The ionic gelation technique for encapsulating entomopathogenic microorganisms can be conducted using sodium alginate²⁶. This technique is relatively easy to implement and does not require high technology, expensive equipment, or skilled labor. Alginates are biodegradable polysaccharides isolated from brown algae, most commonly used in the pharmaceutical and food industries²⁷. Alginate is inexpensive and has numerous applications. Furthermore, it is biocompatible and non-toxic to the environment, animals, and humans²⁸.

A range of information is available on the virulence of EPF against *Ae. aegypti* and host-pathogen interactions^{29–31}. Despite significant progress in studies aiming to develop fungal biopesticides, no biocontrol products based on EPF are commercially available for use against *Ae. aegypti* larvae. Additionally, to the best of our knowledge, there is no information available regarding encapsulation of EPF for the control of *Ae. aegypti*. This study reports the bioprospection of a *Beauveria* isolate found in Rio de Janeiro state, Brazil, with the potential to control *Ae. aegypti* larvae. Furthermore, for the first time, we describe the development of an innovative, eco-friendly, and bio-rational method to control *Ae. aegypti* larvae using encapsulated conidia, which could easily be incorporated into mosquito management strategies.

Results

Morphological and molecular characterization of *Beauveria* sp.

Conidia were round and the conidiophores had a slightly pyriform shape (Fig. 2b- Supplementary Information dataset S2). Also, hyaline hyphae and aseptate structures were observed. The colony surface had a smooth powdery texture with a color ranging from white to cream, with contraction furrows (Fig. 2c- Supplementary Information dataset S2). The inverse side of the colony usually presented yellowish pigmentation (Fig. 2d- Supplementary Information dataset S2). Identification based on sequencing of the ITS region (ITS1 and ITS4) showed 99% homology to *B. bassiana* sequences obtained from Genbank. This isolate was denominated as LPP 143 and deposited in Genbank under accession number PQ144784.

Bioassay results

Initial virulence bioassay

Upon transferring larvae to pure water after initial exposure to different conidial suspensions, low conidial concentrations (approximately 1×10^3 conidia mL⁻¹) were observed due to carry-over with the larvae, however, this low concentration did not interfere with the results.

On the seventh day of evaluations, concentrations of 1×10^8 conidia mL⁻¹ and 1×10^7 conidia mL⁻¹ resulted in 7% larval survival ($S_{50} = 3$ days; $\chi^2 = 468.0$; $P < 0.0001$ vs. control) and 40% ($S_{50} = 5$ days; $\chi^2 = 196.0$; $P < 0.0001$ vs. control), respectively (Fig. 1). For 1×10^6 conidia mL⁻¹, 95% of the larvae survived ($S_{50} =$ undefined; $\chi^2 = 4.585$; $P = 0.0323$ vs. control). Therefore, a concentration of 1×10^8 conidia mL⁻¹ was chosen for all further experiments.

Viability of encapsulated conidia

Regardless of the concentration used, neither alginate nor CaCl₂ negatively affected conidial germination rates when compared with non-encapsulated conidia. Germination rates were 99% for both treatments (Fig. 1. supplementary material S1).

The larvicidal efficacy of encapsulated *Beauveria bassiana* is modulated by the concentration of calcium chloride

Over a seven-day exposure period, conidia encapsulated using 0.01 M CaCl₂ reduced larvae survival to a lesser extent than when encapsulated in CaCl₂ at 0.009 M or 0.008 M (Fig. 2). Furthermore, exposure of larvae to both F and F + B resulted in a reduction in survival (Fig. 2) when compared to their respective control groups. However, conidia encapsulated without bird food were more virulent against larvae than when capsules containing conidia + bird food were tested (Fig. 2). Within seven days, F and F + B with 0.01 M CaCl₂ reduced survival by 30% and 22%, respectively. Regarding CaCl₂ at 0.009 M, the encapsulated fungus alone or associated with food (F + B) reduced larvae survival to 51% and 58% (Fig. 2b). Larval survival when exposed to capsules with 0.008 M CaCl₂, at seven days, was 43% for F, and 54% for F + B, respectively (Fig. 2c).

Differences in virulence for encapsulated conidia with 0.01, 0.009, and 0.008 M CaCl₂ were also compared. Here, conidia encapsulated using 0.009 M or 0.008 M CaCl₂ were more effective against larvae than when encapsulated using 0.01 M CaCl₂ ($P < 0.0001$) (Fig. 2d). Similarly, encapsulated conidia with bird food using 0.009 M or 0.008 M CaCl₂ were also more effective ($P < 0.0001$) against larvae than F + B [0.01 M] (Fig. 2e).

Fungal delivery

On the first day of this experiment, the conidial concentration in the water ranged from 1×10^6 conidia mL⁻¹ to 2.5×10^6 conidia mL⁻¹ for capsules with 0.01 M, 0.009 M, and 0.008 M CaCl₂ ($P \geq 0.05$; Fig. 3). After seven days, the fungal concentration was estimated to be 2.5×10^6 conidia mL⁻¹ for water with capsules formed using 0.01 M, and 5×10^6 conidia mL⁻¹ for capsules formed using 0.009 M CaCl₂. However, water samples with capsules formed using 0.008 M CaCl₂ showed an increase in fungal concentration (approximately 1.5×10^7 conidia mL⁻¹) when compared with the other types of capsules ($P < 0.0001$; Fig. 3).

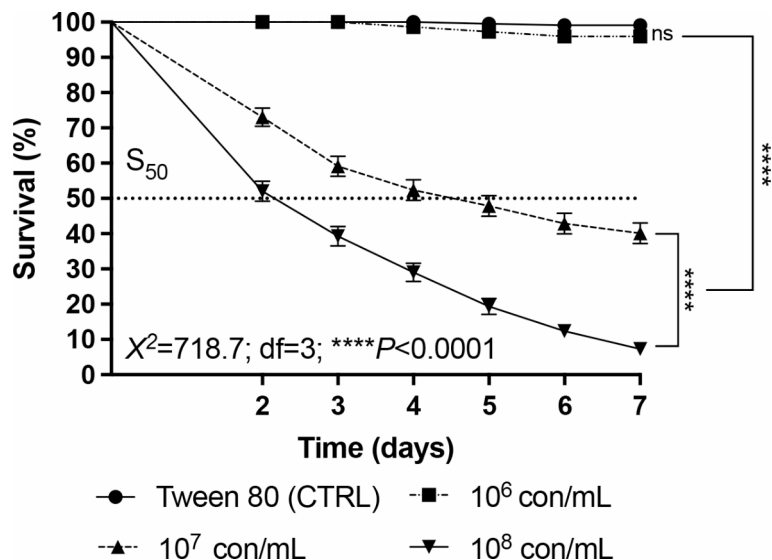


Fig. 1. *Aedes aegypti* survival following exposure to *Beauveria bassiana* conidial suspensions at three different concentrations. Larvae were exposed to conidial suspension for two days. After 48 h, larvae were transferred to pure sterile tap water. The survival was monitored then daily for the next five days. Log-rank test provided the statistical comparison (P values ≤ 0.05), and chi-square values (χ^2). Note: degrees of freedom (df); CTRL = control; con/mL = conidia mL⁻¹. ns = not significant.

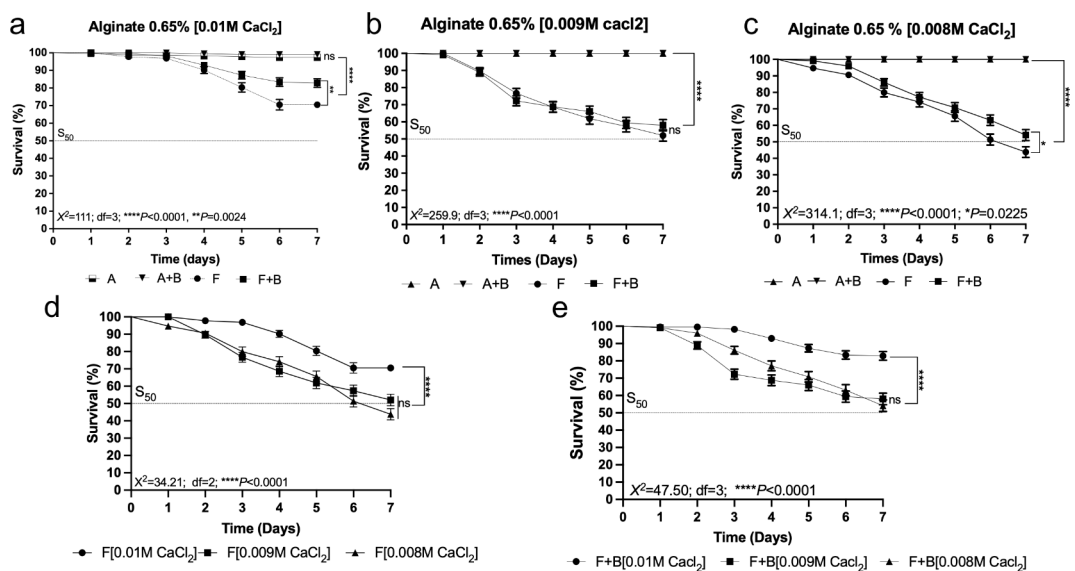


Fig. 2. Survival curves of *Aedes aegypti* larvae exposed to encapsulated *Beauveria bassiana* conidia using 0.65% alginate and different concentrations of calcium chloride (0.01 M, 0.009 M, or 0.008 M). Each group of larvae was exposed to capsules produced using 0.01 M, 0.009 M, or 0.008 M CaCl₂ for seven days. Survival was monitored daily for seven days. The log-rank test provided statistical comparisons (P values ≤ 0.05), and chi-square values (χ^2). Note: degrees of freedom (df), ns = not significant; F = capsules with conidia; F + B = capsules with conidia and bird food; A = capsules produced with pure alginate (control group); A + B = alginate and bird food capsules (extra control group).

In vitro efficacy of mixtures of capsules with different calcium chloride concentrations

Mixtures of encapsulated conidia with different CaCl₂ concentrations significantly reduced larvae survival ($\chi^2 = 317.7$, df = 3, $P < 0.0001$). Within seven days, only 26% of the larvae survived exposure to capsules with fungus alone (MF), whereas 53% of the larvae survived after being exposed to encapsulated conidia with bird food (MF + B) (Fig. 4).

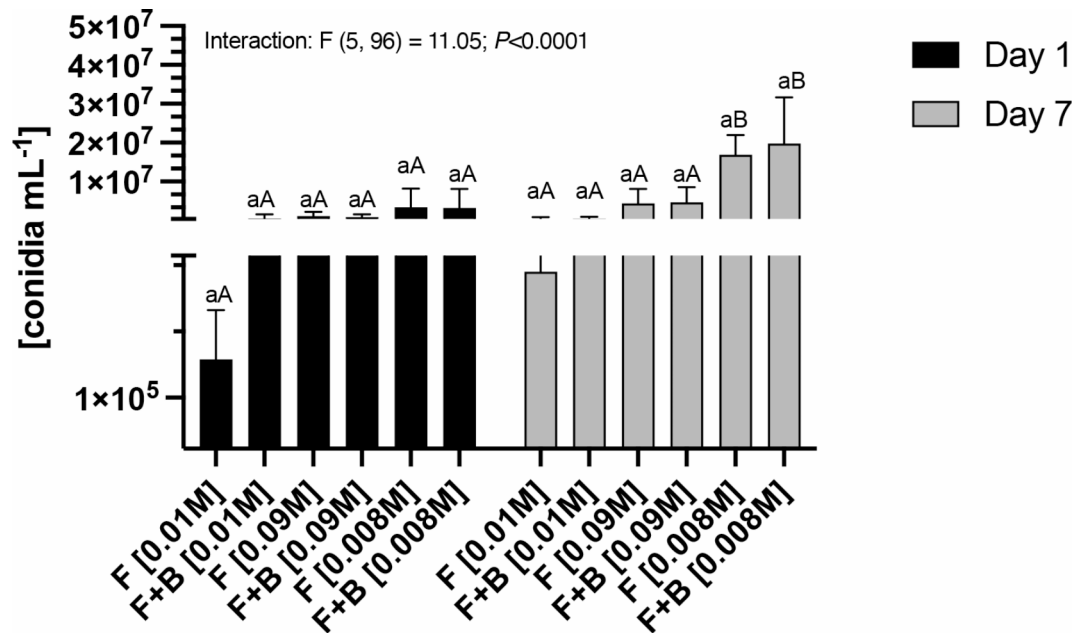


Fig. 3. Amount of conidia released from alginate capsules into the surrounding water. Quantification was performed on the first and seventh days of the experiment. The same lowercase letters indicated no significant differences ($P \geq 0.05$) when comparing different groups at the same time. Results followed by the same capital letters were not statistically different ($P \geq 0.05$) over time (day 1 to day 7). F = capsules with conidia; F + B = capsules with conidia and bird food.

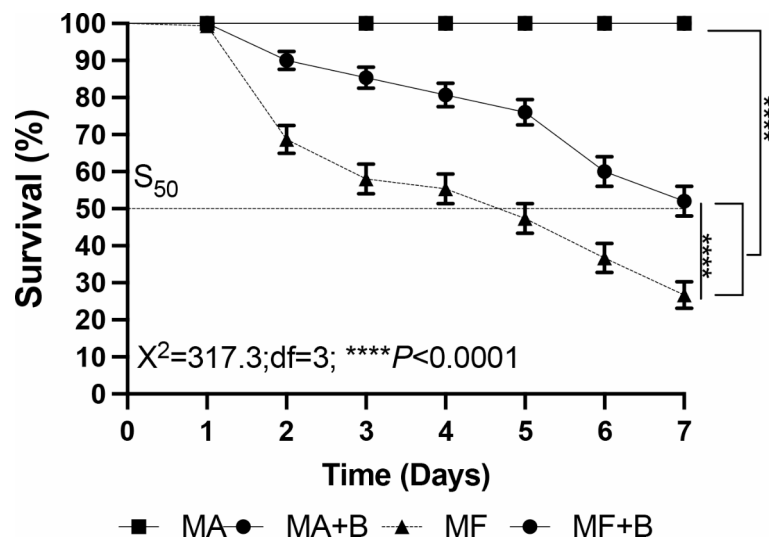


Fig. 4. Survival of *Aedes aegypti* larvae exposed to *Beauveria bassiana* encapsulated using 0.65% alginate with different concentrations of calcium chloride (0.01 M, 0.009 M, or 0.008 M). Each group of larvae was exposed to a mixture of capsules with 0.01 M, 0.009 M, or 0.008 M CaCl_2 for seven days. Survival was monitored daily for seven days. The log-rank test provided statistical comparisons ($P \leq 0.05$), and chi-square (χ^2) values. Note: degrees of freedom (df); ns = not significant; MF = capsules with conidia; MF + B = mixtures of capsules with conidia + bird food; MA = mixtures of capsules of pure alginate (control group); MA + B = mixtures of capsules of alginate + bird food (control group).

To assess the fungal viability after ingestion, an experiment was conducted using larvae that survived exposure to these capsules. Conidia ingested by larvae were 99% viable with no statistically significant difference ($P \geq 0.05$) seen when compared with non-ingested conidia obtained from fungal suspensions.

Encapsulated conidia are effective against larvae under semi-field conditions

Under semi-field conditions, encapsulated conidia were virulent against mosquito larvae ($\chi^2 = 774.9$, $df = 3$, $P < 0.0001$). Over a seven-day observation period, larval survival rates were 35% ($S_{50} = 6$ days) for larvae exposed to encapsulated conidia without diet, and 80% survival ($S_{50} = \text{undefined}$) was observed for larvae exposed to the encapsulated fungus in association with diet (Fig. 5). The climatic conditions for all repetitions, during the study period, were an average temperature of 25.6 °C (range 22–33.5 °C) and an average relative humidity of 85.8% (range 53.5 – 99.6%) (for complete datalogger results see Supplementary Information dataset S3).

Histopathological analysis of larvae exposed to capsules

In both control groups: A (Fig. 6a and a.1) and A + B (Fig. 6b and b.1), all of the larvae had healthy intestines. Well-preserved enterocytes were observed with simple cuboidal epithelium forms without degenerated cytoplasm. No nuclear retraction, and a preserved brush border were observed. Alginate (ag) particles were observed inside the midgut lumen (Fig. 6a and b). Furthermore, no peritrophic membranes were observed.

Fungal propagules were not observed infecting the larvae via cuticle penetration. However, interactions in the midgut were observed. Larvae exposed to capsules with fungus only had abnormal midguts with degenerated enterocytes (2 out of 3 samples), and disrupted brush borders (2 out of 3 samples). An intense production of extracellular material was observed in the ectoperitrophic space for all samples. Peritrophic membranes (empty arrow: Figures 7 and 8) were observed in all of the samples. Two out of three samples showed enterocytes with a columnar shape (Fig. 7b). Furthermore, ungerminated conidia were observed in all samples as well as germinating conidia (1 out of 3 samples).

All 3 samples from larvae exposed to F + B showed enterocytes with columnar cells. No degenerated cytoplasm was observed and no disrupted brush borders (bb: Fig. 8a and a.1). However, the enterocytes demonstrated nuclear (n) retraction (Fig. 8a, b, and c) in all samples. Production of extracellular material (ecm) in the ectoperitrophic space (ecp) was also observed in all samples. Furthermore, the presence of peritrophic membranes were observed in all of the samples. Germinating conidia (arrows) were observed in all samples (Fig. 8a, b, c and c.1).

Discussion

For the first time, alginate-encapsulated conidia from a new Brazilian *B. bassiana* isolate were tested against mosquito larvae. A series of experiments to investigate virulence and host-pathogen interactions demonstrated the efficacy of this method under both laboratory and semi-field conditions against *Ae. aegypti* larvae.

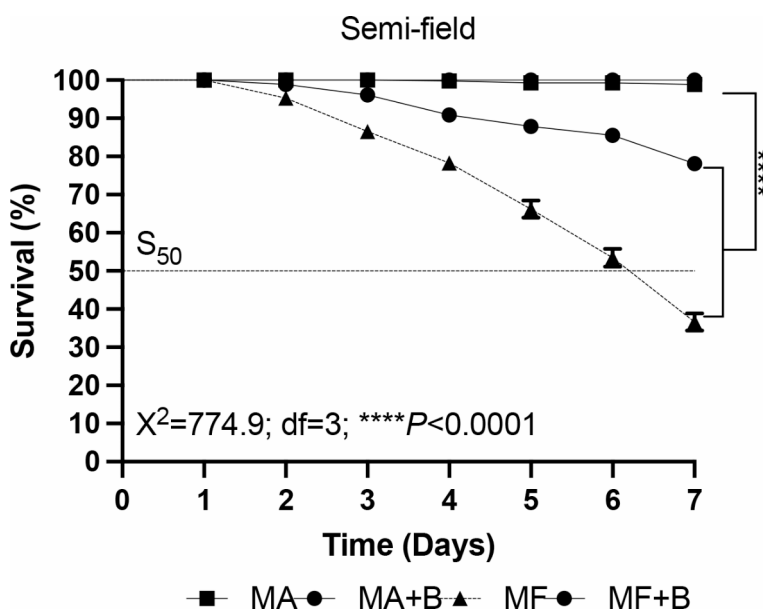


Fig. 5. *Aedes aegypti* larval survival rates following exposure to *Beauveria bassiana* capsules with 0.65% alginate and different concentrations of calcium chloride (0.01 M, 0.009 M, or 0.008 M). Each group of larvae was exposed to 20 g of capsules with 0.01 M, 0.009 M, and 0.008 M CaCl_2 , totaling 60 g of capsules per liter of water in each container. Survival was monitored daily for seven days under semi-field conditions. The log-rank test was used for statistical comparison (P values ≤ 0.05), and calculating the chi-square value (χ^2). Note: degrees of freedom (df); MF = mixtures of capsules with conidia; MF + B = mixtures of capsules with conidia and diet; MA = mixtures of pure alginate capsules (control group); MA + B = mixtures of capsules with alginate and diet (second control group).

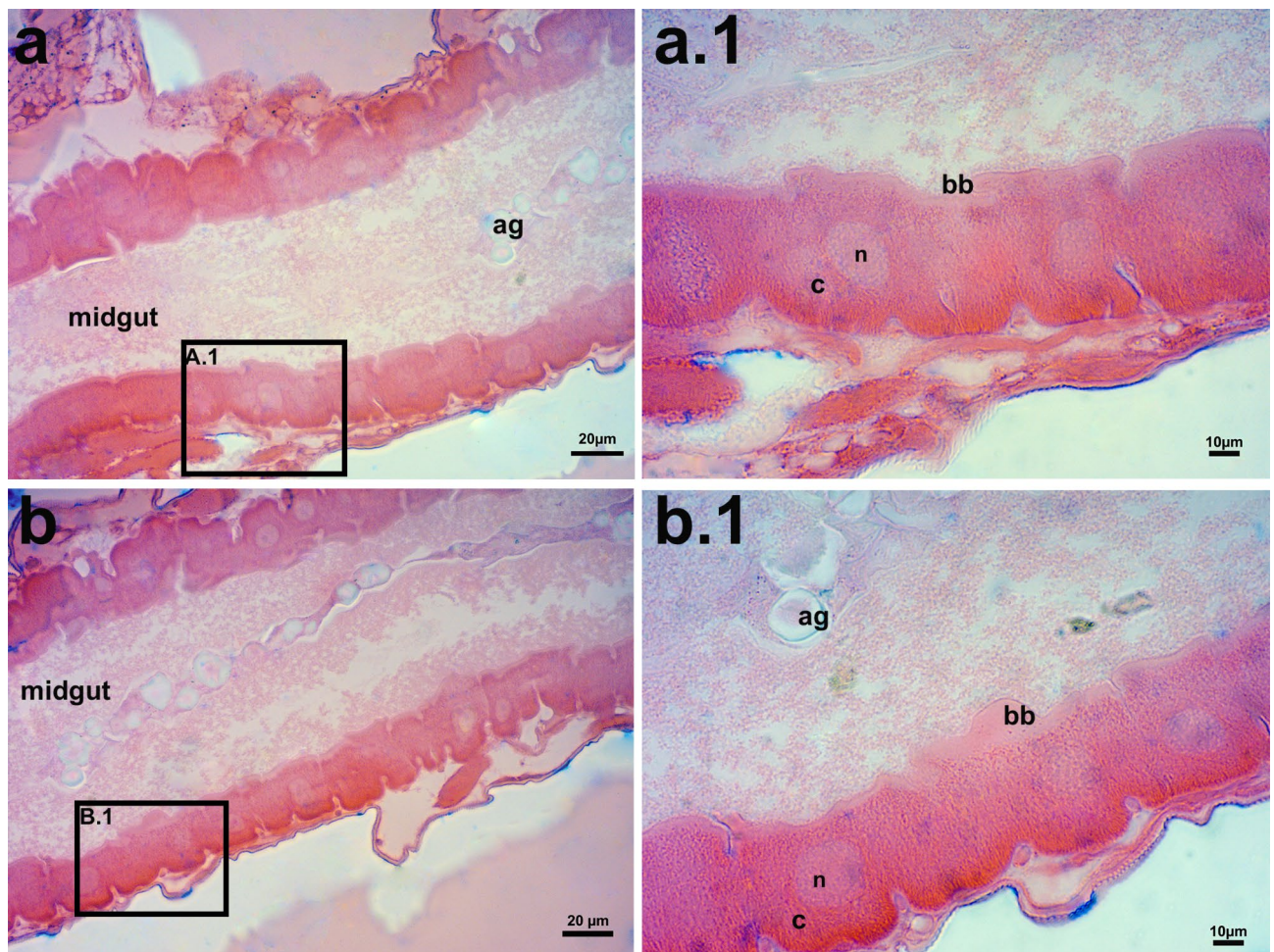


Fig. 6. *Aedes aegypti* larval midgut following exposure to alginate capsules (A and A.1), or alginate combined with bird food (B and B.1). Regardless of the treatment, there were no abnormal observations. Specimens demonstrated healthy cuboidal epithelial cells (enterocytes) with normal brush borders (bb), nucleus (n) with no retraction or degenerated cytoplasm (c) ag = alginate.

The macro and micromorphological characteristics of this fungus were used to identify it as *B. bassiana* (designated as LPP 143), and molecular biology was used to confirm the species. *Beauveria bassiana* LPP 143, recently isolated from the coffee berry borer *Hypothenemus hampei* (Coleoptera: Curculionidae), was shown to have the potential to control *Ae. aegypti* larvae. In virulence bioassays, 10^8 conidia mL^{-1} killed 50% of larvae in 3 days and approximately 100% in seven days, corroborating with Kirsch and Tay³², Mendonça et al.³³, Tawidian et al.³⁴, who also reported the efficacy of *B. bassiana* isolates against *Aedes* mosquitoes. Other isolates of filamentous fungi have been reported to be effective against insects of public health concern^{35,36}, and this new isolate is a promising candidate for further studies.

Here, conidia showed high percentage of viability after encapsulation. According to Rodrigues et al.³⁷ the ionic gelation technique using alginate might benefit EPF by maintaining fungal viability and virulence, also reducing the impact of deleterious abiotic factors such as ultraviolet radiation and high temperatures. For example, Meirelles et al.²⁵ showed that encapsulated *Metarhizium* were more resistant to UV-B radiation (6.0 and 8.0 kJ m^{-2}) and heat (42 °C for 2–6 h), maintaining a high percentage of conidial germination when compared to non-encapsulated conidia.

Conidia from LPP 143 encapsulated using 0.65% alginate effectively reduced larvae survival. Before testing 0.65% alginate, 2% and 1% alginate (data not shown) were used to form capsules with 0.1 M CaCl_2 according to Meirelles et al.²⁵. These authors applied 30, 60 or 90 mg aliquots of fungal capsules onto sterile soil sample surfaces and tested these against engorged female cattle ticks *Rhipicephalus microplus*. After placing the ticks on the soil samples, the authors observed a reduction of the oviposition rates (30.3% for 3% alginate, and 57.6% for 2% alginate) and an increase in female mortality five days after exposure of the ticks to the capsules. However, in an aquatic environment, conidia were not efficiently released from capsules when using high concentrations of CaCl_2 and were therefore ineffective against mosquito larvae. Subsequent tests, combining different concentrations of CaCl_2 and alginate were also ineffective. For example, formulations with less than 0.65% alginate did not effectively form capsules during the dripping process. However, when 0.65% alginate was dripped into CaCl_2 at 0.01 M, 0.009 M, and 0.008 M these combinations yielded well formed capsules.

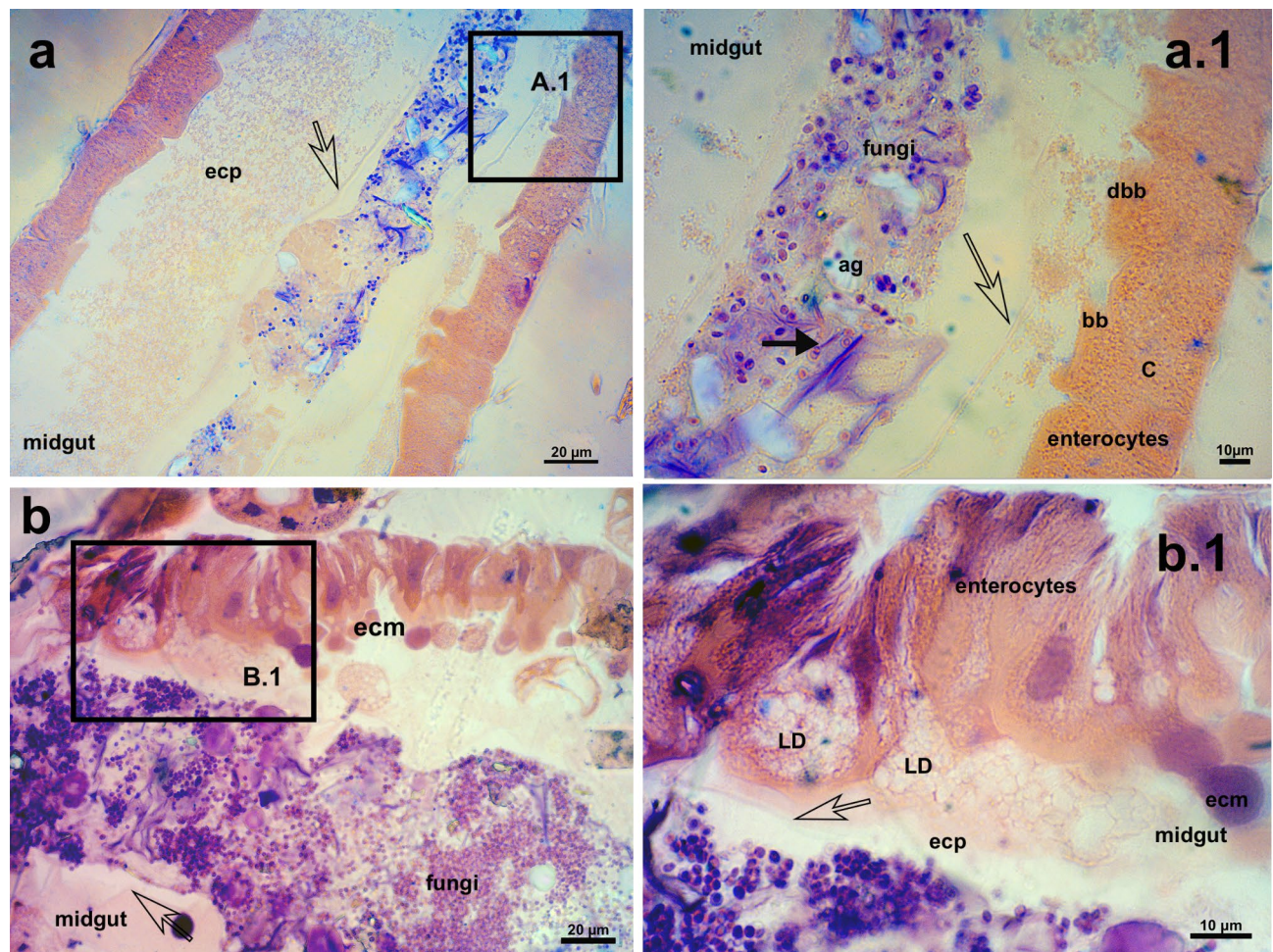


Fig. 7. *Aedes aegypti* larvae midgut sections following exposure to capsules with conidia only (A; A.1; B and B.1). Abnormal morphology was observed with degenerated enterocytes (A.1) and disrupted brush borders (dbb). Peritrophic membrane (empty arrow), production of extracellular material (ecm), lipid droplets (LD), and the presence of *Beauveria bassiana* propagules in the midgut lumen were observed. ecp = ectoperitrophic space; ag = alginate.

Interestingly, according to Li et al.³⁸ higher concentrations of CaCl_2 increase the cross-link density of the alginate beads, and their swelling degree and consequently, decrease the interaction with the water.

After encapsulation in different CaCl_2 concentrations, the next experiment investigated the amount of conidia released into the water. Capsules produced with 0.008 M CaCl_2 released significant numbers of conidia into the water, reaching a concentration of 10^7 conidia mL^{-1} over seven days. Notably, in the initial in vitro bioassays using capsules, larval survival was approximately 50%, consistent with our first bioassay using a fungal suspension of 1×10^7 conidia mL^{-1} . On the other hand, capsules with 0.01 M CaCl_2 (i.e., higher density) released lower amounts of conidia, resulting in higher survival rates when compared to the other capsule formulations. Accordingly, capsules with 0.01 M CaCl_2 killed 30% and 20% of the larvae in the F and F + B groups, respectively. Capsules with 0.01 M or 0.009 M CaCl_2 released approximately 10^6 conidia mL^{-1} into the water. The higher efficacy of encapsulated conidia when compared to pure fungal suspensions of 10^6 conidia mL^{-1} , was expected since the larvae were exposed to fungal suspensions for only 24 h, whereas exposure to the capsules occurred over seven days.

No information is available regarding the use of EPF-alginate capsules against *Aedes*. However, some studies have investigated the use of alginate capsules with other microorganisms or active ingredients for mosquito larval control. For instance, when *Ae. aegypti* females were exposed to clove basil *Ocimum gratissimum* (L.) oil-alginate capsules, a 100% oviposition deterrence was observed on the third day of the experiment³⁹. These studies indicate the potential of alginate capsules in delivering biocontrol agents for mosquito population management, although specific applications for entomopathogenic fungi against *Aedes* remain to be investigated. Also, Prabakaran and Hoti⁴⁰ tested different multivalent counterions, including calcium chloride and zinc sulfate to encapsulate *Bacillus thuringiensis* var. *israelensis* (Bti) for controlling *Culex quinquefasciatus* (Diptera: Culicidae). They showed that 98% of larvae were killed following exposure to zinc alginate-Bti capsules.

Here, we demonstrated that virulence was inversely proportional to CaCl_2 concentration i.e., with decreasing CaCl_2 concentrations, the capsules released more conidia into the water and thus killed more larvae in less time.

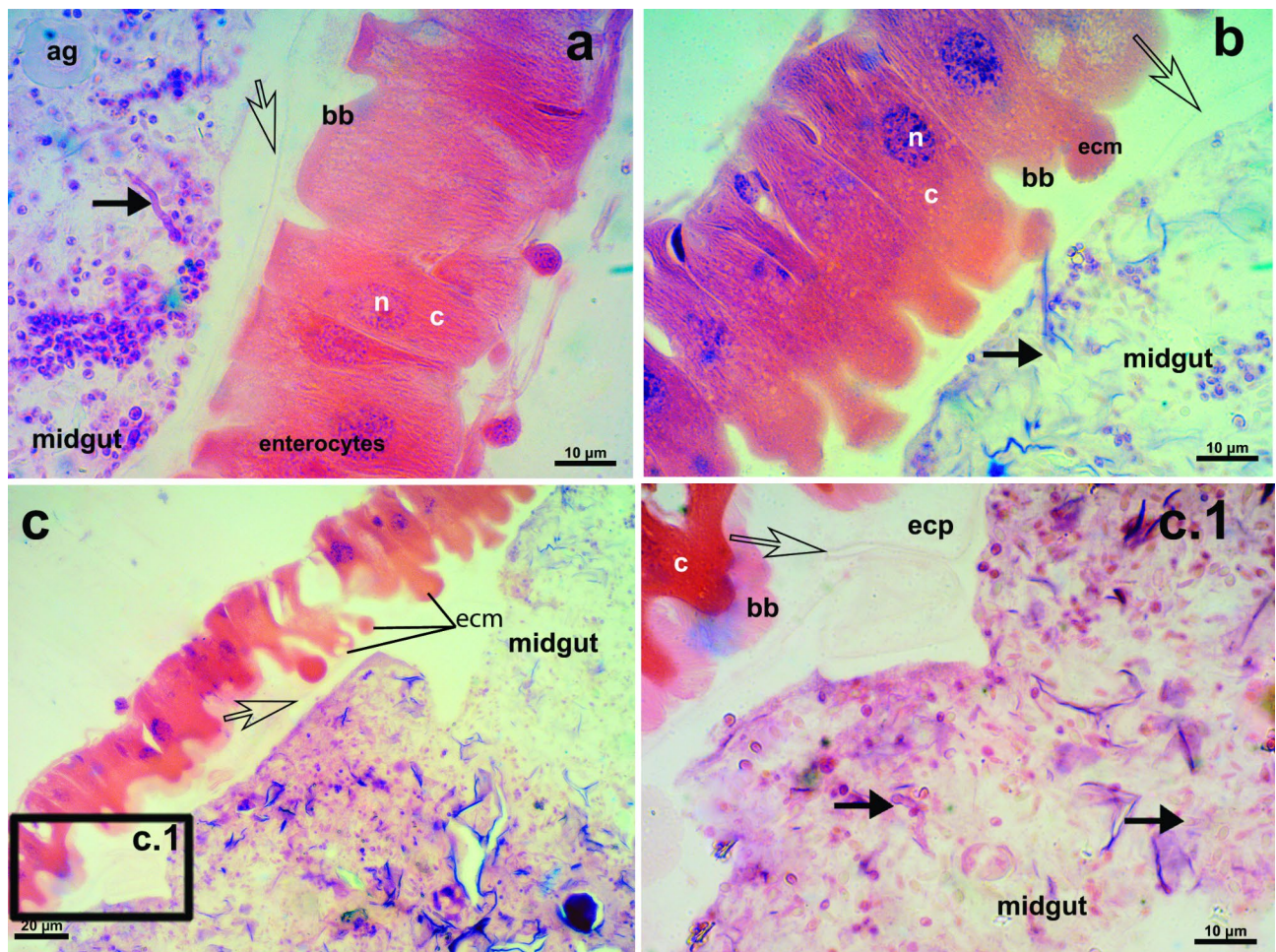


Fig. 8. *Aedes aegypti* larval midguts following exposure to encapsulated conidia combined with bird food (A, B, C, and C.1). The midguts showed columnar enterocytes with homogenous cytoplasm (c) but with nuclear (n) retraction. Brush borders (bb) were observed without degeneration. Furthermore, the presence of peritrophic membranes (empty arrow), the production of extracellular material (ecm), and germinated conidia (arrows) were observed.

The capsules with the lowest concentrations of CaCl_2 also disintegrated the fastest. Although an experiment was not conducted to determine how long it takes for a capsule to completely disintegrate in water, 0.008 M and 0.009 M CaCl_2 capsules retained their original form for an average of 24, and 48 h, respectively, while capsules with 0.01 M CaCl_2 maintained their format throughout the entire experiment.

The next in vitro bioassay was conducted to evaluate larval susceptibility when using mixtures of capsules of different characteristics: 0.01 M CaCl_2 (slow-release rate), 0.009 M CaCl_2 (medium-release rate), and 0.008 M CaCl_2 (quick-release rate). For this bioassay, 1.5 g of each capsule were used, totaling 4.5 g per cup (i.e., the same total amount as used in the previous bioassay). Fungus-only capsules killed more larvae than capsules combined with bird food. Furthermore, under semi-field conditions, capsules maintained their virulence against larvae and once again capsules without bird food were more effective than capsules with bird food. Even when temperatures fluctuated, capsules showed promising results in semi-field conditions. Further experiments will focus on shelf-life and persistence of the capsules.

We did not conduct experiments on mosquito behavior; however, larvae were attracted to capsules (supplementary material video 1). Curiously, the capsules which had been formed with the inclusion of diet were less virulent than the fungus-only capsules. Histological observation revealed smaller amounts of propagules in the midgut of larvae exposed to capsules with bird food when compared to larvae exposed to fungus-only capsules. Thus, we suggest that the bird food occupies physical space that could otherwise be taken up by fungal propagules, resulting in lower numbers of conidia released into the environment.

Here, conidia did not infect the insect via cuticle penetration, corroborating the results seen by Butt et al.⁴¹. Greenfield et al.⁴² reported the absence of long-chain hydrocarbons in the cuticle of mosquito larvae that are responsible for stimulating fungal adherence and development on the integument. Notably, Ramirez et al.⁴³ reported an increase in the midgut microbiota when *Ae. aegypti* adults were exposed to EPF, resulting in a deregulation of midgut homeostasis, assisting the fungus to successfully kill the mosquito. Furthermore, Wei et al.⁴⁴ reported a downregulation of antimicrobial peptides and dual oxidase in *Ae. aegypti* adults exposed

to oosporein toxin produced by *B. bassiana*. These alterations caused the translocation of *Serratia marcescens* bacteria from the midgut into the hemocoel, killing the mosquito.

In our experiments, histopathological analysis revealed germinating propagules in the gut lumen, as also reported by Noskov et al.²¹, who detected *Metarhizium robertsii* inside larval hemocoel after accumulation in the gut of the immature stage. In addition, sections exhibited enterocytes with abnormal aspects such as degenerated cytoplasm and disrupted brush borders. Bitencourt et al.⁴⁵ reported similar results in *Aedes* larvae exposed to *Metarhizium anisopliae* CG153. The degradation of midgut epithelia is well documented in larvae infected by Bti which produces Cry and Cyt proteins that create pores in the gut⁴⁶. EPF produce digestive enzymes including lipases, chitinases, and proteases which help the fungus adhere to and degrade the insect cuticle¹⁵. *Beauveria* sp. is known to produce a range of secondary metabolites, including beauvericin, beauverolides, tenellin, and oosporein, which aid in the infection process⁴⁷. Among these, beauvericin, a cyclic hexadepsipeptide toxin, is particularly well-studied and has the ability to increase ion permeability in biological membranes⁴⁸. Jow et al.⁴⁹ reported that beauvericin induces cell death through an apoptotic pathway, decreasing mitochondrial transmembrane potential, increasing cytochrome c, and activating caspase-3. Thus, we propose that in an attempt to infect the larvae, *B. bassiana* LPP 143 conidia might release enzymes and toxins in the gut, affecting enterocytes and leading to larval death. Furthermore, although the midgut is composed of regenerative epithelial cells⁵⁰, the fungal attack might occur faster than the cell regeneration process. Future experiments will focus on investigating the production of toxins in the digestive tract.

The observation of germinated conidia in the larval digestive tract led us to conduct an experiment to investigate the viability of conidia present in the gut lumen. Butt et al. (2003) reported that *Metarhizium* sp. fails to germinate after passing through the gut. Additionally, Dillon and Charnley⁵¹ reported that ingested conidia showed a significant reduction in germination. These authors observed a 73.4% decrease in conidial viability of *Metarhizium anisopliae* in the gut lumen of the desert locust (*Schistocerca gregaria*). Interestingly, they attributed this fungicidal effect to the gut microbiota, rather than factors such as pH, osmotic pressure, or the insect's diet. In contrast, our study showed that the conidia recovered from the gut remained viable in germination assays.

Mosquito gut cells are capable of producing peritrophic membranes and an array of digestive enzymes. The peritrophic membrane represents a physical barrier against pathogens and the gut epithelium is responsible for acquiring and storing nutrients⁵⁰. The histopathological observations revealed the presence of the peritrophic membrane only in groups treated with fungus, suggesting a rapid production of these barriers to protect against EPF. In contrast, the absence of peritrophic membranes in the control groups might be due to the lack of a stimulus for membrane formation, as the capsules contained only alginate and food (i.e., without the pathogen).

Following exposure to capsules-containing conidia, it was observed that larval locomotor capacity and their ability to reach the water surface to breathe was reduced. Butt et al.⁴¹ observed that conidia blocked the respiratory syphon of the larvae which may also have contributed to larval death.

Conclusion

This work successfully demonstrated a new approach for the use of EPF to control aquatic mosquito larvae, that was also effective in semi-field conditions. Alginate capsules formulated with conidia exhibited a good release rate and maintained their larvicidal activity both under laboratory and semi-field conditions. The new *B. bassiana* isolate tested here is a promising candidate for future studies, and its ability to damage the digestive tract and kill larvae demonstrates the multifaceted aspects of EPF infection. Entomopathogenic fungi have the ability to infect terrestrial and aquatic insects via the cuticle and/or digestive tract. Therefore, new biological products based on EPF could be incorporated in future integrated vector control strategies.

Materials and methods

Insect collection and maintenance

Aedes aegypti eggs (collected around the university campus) were provided by the Grupo de Patologia de Insetos da Universidade Estadual do Norte Fluminense (UENF, Brazil). The eggs were collected using ovitraps (black plastic plant pots with wooden strips), according to Carolino et al.¹³. Briefly, the ovitraps with 300 mL of tap water were placed outdoors under a covered veranda located at the university insectary, to attract adult females to oviposit. After 5 days, the strips were collected and dried at room temperature for 24 h. The presence of eggs was confirmed using a dissecting microscope. To stimulate larval hatching, the strips were submerged in plastic trays containing water. Larvae were separated (100 larvae 1 L⁻¹ of water) and fed with granulated sterile bird food (Alcon, Santa Catarina, Brazil) on a daily basis. Pupae were transferred to plastic cups containing water (40 pupae/300 mL), and covered with mesh and sealed with a rubber band. Two-three day old adults were transferred to cages and fed *ad libitum* with 10% sucrose solution. To obtain eggs for the next generation of larvae (F₁), adults were artificially fed with rabbit blood⁵². The eggs laid by blood-fed females were collected from filter paper used to line the breeding cages. To stimulate larval hatching, the eggs were maintained in tap water (2 L) under controlled conditions [temperature 27 ± 1 °C; relative humidity-RH ≥ 80%] with a 12-h light/dark cycle. Larvae were maintained in trays and were fed daily with 0.5 g L⁻¹ granulated sterile bird food. For each experiment/repetition eggs were collected in the field and the above process carried out.

Fungus: isolation and maintenance

The *Beauveria* sp. isolate used here was originally found infecting an adult coffee berry borer (*Hypothenemus hampei*), collected at the São José farm (Bom Jardim, Rio de Janeiro, Brazil) (geographical coordinates – 21.76192835451863 S, -42.07171676656515 W, Isolation date- 04/19/2022). First, the coffee bean containing the infected coffee berry borer was incubated in a humidity chamber for seven days under controlled conditions [25 ± 1 °C; RH ≥ 90%] to improve fungal sporulation⁵³. Next, the fungus was inoculated, by using a streak plate

method, onto potato dextrose agar (PDA- HiMedia Laboratories, PA, USA) containing 0.05% chloramphenicol to prevent bacterial contamination during the purification process. After 15 days, a monospore technique⁵⁴ was conducted to obtain a pure fungal culture. Once purified, the isolate was then replated onto PDA, without antibiotics^{53,54}. The isolate was maintained under controlled conditions [25 ± 1 °C; RH $\geq 80\%$] and incorporated into the Fungal Collection of the Insect Pathology Group UENF under accession code LPP 143.

Morphological and molecular characterization

Macromorphological observations were assessed using the inoculation point technique⁵⁵ from 5 Petri dishes ($N=15$) as described previously. For micromorphological characteristics, the fungus was cultivated on PDA between a glass slide and coverslip under controlled conditions [25 ± 1 °C; RH $\geq 80\%$]³⁴. After five days the fungus was stained with lactophenol blue and observed using light microscopy (Axioplan, Zeiss, Jena, Germany) at 1000× magnification⁵⁶ and images were captured using an AxioCam MR3 camera (Zeiss, Jena, Germany).

For molecular analysis, 100 mg of conidia were removed from PDA culture media, and DNA was extracted using a Quick-DNA™ Fungal/Bacteria miniprep kit following the manufacturer's instructions (Zymo Research, CA, USA). Subsequently, the extracted DNA was quantified using a NanoDrop 2000 spectrophotometer (Thermo Scientific, MA, USA). Internal transcribed spacers (ITS) universal primers were used for fungal identification⁵⁷. Primer details and Polymerase Chain Reaction (PCRs) conditions can be found in supplementary material 1.

The sequencing primers were the same as those used for PCR. The sequence contigs were assembled and manually edited using Geneious v.8.1.9 (Biomatters, CA, USA). The BLASTn search tool was used to establish preliminary fungal identification.

Preparation of fungal suspensions

Conidia were harvested and suspended in sterile 0.03% Tween-80 in dechlorinated tap water (v/v). The suspensions were homogenized in a vortex mixer for one minute. The conidial concentration was adjusted to 1×10^8 conidia mL⁻¹ using a Neubauer chamber (HBG Blood Counting chambers, Imp. Neubauer. Supplier: HBG, Germany). Concentrations of 1×10^7 , and 1×10^6 conidia mL⁻¹ were obtained by serial dilution⁵³. These fungal suspensions were prepared for the initial screening trials against mosquito larvae.

Fungal encapsulation

Encapsulated conidia were prepared using an ionic gelation encapsulation technique²⁵. Briefly, a 0.65% (w/v-0.0065 g mL⁻¹) solution of sodium alginate (Proquimios, Rio de Janeiro, Brazil) was prepared by dissolving the polymer in pure distilled water. The concentration of sodium alginate was chosen based on preliminary tests conducted to assess capsule formation and efficiency of fungal delivery. Before associating the fungus with alginate, the conidial concentration was determined by weighing the dry spores. The spores were suspended in 0.03% Tween 80 and quantified using a Neubauer chamber. To achieve a concentration of 1×10^8 conidia mL⁻¹, 0.0187 g of conidia was added to each mL alginate solution. The formulation was homogenized using a vortex mixer (VTX-2500-Biomixer Minas Gerais, Brazil) for three minutes and then filtered with a sterile gauze. The formulation was then dripped into either 0.01 M, 0.009 M, or 0.008 M (w/v) calcium chloride (CaCl₂) (Sigma-Aldrich, MO, USA) solution using a perfusion pump (Masterflex L/S™, Cole Palmer, CA, USA) at a flow rate of 1 mL min⁻¹. This process was used to form capsules of different consistencies. Next, the capsules were washed in sterile distilled water to remove excess CaCl₂ and then used immediately in the bioassays. Capsules were also prepared by adding 0.5 g L⁻¹ of macerated bird food (Curió, Alcon Club, Santa Catarina, Brazil) to the conidial suspensions/formulation. Sterile bird food was added with the aim of testing it as a possible larval attractant. The treatment groups were denominated as described in Table 1.

Conidial viability following encapsulation

Conidial viability was assessed to evaluate the possible impact of the encapsulation process on germination. Here, 1 g of capsules was prepared as previously described. After preparation, the capsules were dissolved in 10 mL of 0.5% (w/v) sodium citrate (Neon LTDA, SP, Brazil) plus 0.03% (v/v) Tween 80⁵⁸. The conidial concentration was verified using a Neubauer chamber and adjusted to 1×10^3 conidia mL⁻¹ by serial dilution. A suspension of non-encapsulated conidia (control) was also prepared to compare germination rates.

Ten microliters of fungal suspension were inoculated into Petri dishes ($n=3$) containing PDA + 0.05% (w/v) chloramphenicol. The Petri dishes were incubated under controlled conditions (25 ± 1 °C; RH $\geq 80\%$). After 24 h, 10 µL of lactophenol cotton blue was added to the plates. The percentage germination was determined using light microscopy. Three hundred conidia were counted to determine germinated (i.e., visible germ tubes) and non-germinated (i.e., no visible germ tube) propagules. The experiment was performed three times using different batches of conidia over three different periods.

Biological assays

Initial virulence screening assay

For survival analyses, 25 (L_{2-3} , $N=75$) *Ae. aegypti* larvae were transferred to 80 mL plastic cups [width (W) 4.5 cm x height (H) 5.6 cm] containing 30 mL of fungal suspensions at 1×10^6 , 1×10^7 , or 1×10^8 conidia mL⁻¹ in Tween 80 (0.01%)⁵⁹. A fungal concentration of 1×10^8 conidia mL⁻¹ was estimated using a Neubauer chamber according to Alves⁵³. Serial dilutions were then conducted to obtain 1×10^7 , and 1×10^6 conidia mL⁻¹⁵³. The control group was exposed to Tween 80 (0.01%) only⁵⁹. The experiment was conducted under controlled conditions (27 ± 1 °C; RH $\geq 80\%$). Forty-eight hours after exposure to fungal suspensions, the insects were carefully transferred to plastic cups containing 100 mL of pure sterile tap water using a plastic Pasteur pipette⁶⁰. To check that conidia were not transferred together with the larvae in large quantities, the presence of fungus in the water was verified using a Neubauer chamber. Larval survival was monitored daily for seven days, and the

Group names	Capsule content
F [0.01 M]	0.65% alginate + 0.01 M CaCl ₂ + conidia
F + B [0.01 M]	0.65% alginate + 0.01 M CaCl ₂ + conidia + bird food
F [0.009 M]	0.65% alginate + 0.009 M CaCl ₂ + conidia
F + B [0.009 M]	0.65% alginate + 0.009 M CaCl ₂ + conidia + bird food
F [0.008 M]	0.65% alginate + 0.008 M CaCl ₂ + conidia
F + B [0.008 M]	0.65% alginate + 0.008 M CaCl ₂ + conidia + bird food.
A [0.01 M]	0.65% alginate + 0.01 M CaCl ₂ (without conidia)
A + B [0.01 M]	0.65% alginate + 0.01 M CaCl ₂ + bird food (without conidia)
A [0.009 M]	0.65% alginate + 0.009 M CaCl ₂ (without conidia)
A + B [0.009 M]	0.65% alginate + 0.009 M CaCl ₂ + bird food (without conidia)
A [0.008 M]	0.65% alginate + 0.008 M CaCl ₂ (without conidia)
A + B [0.008 M]	0.65% alginate + 0.008 M CaCl ₂ + bird food (without conidia)

Table 1. Capsule formulations used against *Aedes aegypti* larvae.

surviving larvae were fed daily with sterile bird food (0.5 g L⁻¹). The experiment was performed three times with different batches of conidia over three different periods.

In vitro virulence of encapsulated *Beauveria* against larvae

Groups of 25 *Ae. aegypti* larvae (L₂₋₃; N=75) were transferred to plastic cups (80 mL; W 4.5 cm x H 5.6 cm) containing 30 mL of sterile tap water. The groups were exposed, separately, to capsules with fungus [0.01 M], fungus + bird diet [0.01 M], fungus [0.009 M], fungus + bird diet [0.009 M], fungus [0.008 M], or fungus + bird diet [0.008 M]. The groups were denominated as described in Table 1. Each group/cup received 4.5 g of capsules. The control group was exposed to capsules without fungus (with or without the addition of bird food to the capsules). The experiment was performed under controlled conditions (27 ± 1 °C; RH ≥ 80%; Light = 12 h: Dark = 12 h). Survival monitoring was conducted as previously described.

On the first and last day of the experiment, 10 µL of water was collected from the cups to estimate the quantity of conidia released into the water from the capsules. The conidial concentration was determined using a Neubauer chamber. The experiment was performed three times with different batches of conidia over three different periods.

In vitro virulence of mixtures of capsules with different consistencies

Capsules with 0.01 M, 0.009 M, and 0.008 M CaCl₂ were prepared separately and then combined (1.5 g of each = 4.5 g in total). The aim was to investigate whether a combination of each type of capsule would remain effective against larvae. Groups of 25 *Ae. aegypti* larvae (L₂₋₃; N=75) were transferred to plastic cups with 30 mL of sterile tap water. Each group received 4.5 g of capsules. The control group was exposed to capsules without fungus (with or without bird food). Survival monitoring was conducted as previously described. Mixtures of capsules were denominated as follows: MF (encapsulated fungus prepared with 0.01 M, 0.009 M, and 0.008 M CaCl₂), MF + B (encapsulated fungus + bird food prepared with 0.01 M, 0.009 M, and 0.008 M CaCl₂); MA (only alginate encapsulated with 0.01 M, 0.009 M, and 0.008 M of CaCl₂); MA + B (encapsulated fungus + bird food prepared at 0.01 M, 0.009 M, and 0.008 M CaCl₂). The experiment was performed three times with different batches of conidia over three different periods.

Efficacy of mixtures of capsules against larvae under semi-field conditions

For the semi-field bioassays, 50 *Ae. aegypti* (L₂₋₃; N=150) larvae were transferred to plastic containers (W 15 cm x H 10 cm) with one liter of sterile tap water and 60 g (calculated in proportion to the in vitro experiments) of capsule mixtures (i.e., 0.01 M + 0.009 + 0.008 M = 20 g of each type totaling 60 g). The control group was exposed to capsules without fungus (with or without bird food). The experiment was carried out under shaded conditions (a covered veranda) at ambient temperatures. Over the test period, a data logger (Escort RH iLog, Escort Data Logging Systems, Switzerland) was used to record the relative humidity and temperature. Survival monitoring was conducted as previously described. The experiment was performed three times with different batches of conidia over three different periods. The experiment was conducted according to WHO guidelines⁶⁰ with modifications to the exposure times.

Histopathological analysis of larvae exposed to a mixture of capsule types

Groups of 10 *Ae. aegypti* larvae (L₂₋₃; N=30) were exposed to mixtures of capsules, as described in Sect. 2.6.3. After 72 h, the larvae (N=3 per group) were fixed in a 4% paraformaldehyde (Sigma-Aldrich, MO, USA) solution for 24 h under refrigeration and dehydrated in an ascending ethanol series (15 min each at 70, 80, 90, 96, and 100% ethanol). The dehydration process with ethanol at 100% was repeated three times for 15 min each⁵⁹. Incubation was conducted for 12 h in historesin (Leica, Heidelberg, Germany) diluted in 100% ethanol (Merck, Darmstadt, Germany) (1:1 v/v).

Next, the samples were incubated in 100% historesin for 24 h, followed by hardener in 100% historesin. A rotary microtome (RM2135 Leica, Wetzlar, Germany) was used to obtain 3 µm thick histological sections.

The samples were mounted on slides and stained with Alcian blue (AB) (Suetam Ltd., São Paulo, Brazil) in combination with periodic acid-Schiff (PAS) (Sigma-Aldrich, MO, USA) as described by Barbosa et al.⁶¹ with modifications. The sections were incubated in AB solution (pH 2.5) for 20 min. Subsequently, for PAS staining, the sections were incubated in a 1% periodic acid solution for 5 min, followed by treatment with Schiff's reagent for 45 min. The slides were covered and sealed with Entellan (Sigma-Aldrich, MO, USA). The sections were examined using a light microscope (Axioplan, Zeiss, Jena, Germany) and images were captured using an AxioCam MR3 camera (Zeiss, Jena, Germany) coupled to the equipment. Histopathological sections were obtained from the entire larval body, focusing primarily on the cuticle and the gut.

Fungal viability after ingestion by the larvae

To verify if conidia from the capsules were still viable after larval ingestion, groups of 25 larvae (L_{2-3} , $N=75$) were exposed to encapsulated conidia, as described in the section regarding in vitro efficacy of capsule consistency mixtures. After 72 h, the guts of three *Ae. aegypti* larvae ($N=6$ per group) were collected using n°5 tweezers. The guts were pooled and transferred to 100 μ L Tween 80 (0.03%). The samples were macerated with a pestle and vortexed twice for one minute. The presence of conidia was confirmed, quantified, and adjusted to 1×10^3 conidia mL^{-1} using a Neubauer chamber. Ten microliters of the suspension were inoculated in the center of PDA solid media ($N=3$). After 24 h, germination was verified as previously described (conidial viability following encapsulation). The experiment was performed three times with different batches of conidia over three different periods.

Statistical analysis

The survival curves were plotted using Kaplan-Meier analysis and compared using the Log-rank test (5% significance level; $P \leq 0.05$). Furthermore, Log-rank was used to calculate the median survival time (S_{50}). Data for germination and fungal release into the water were submitted to the Shapiro-Wilk test for the distribution of normality. Parametric data was analyzed using a one-way ANOVA or Two-way ANOVA test, followed by Tukey's multiple comparison test. All data analysis was carried out using GraphPad Prism (v8.00) software (GraphPad Software, Boston, MA, United States).

Data availability

The datasets used and/or analysed during the current study available from the corresponding author on reasonable request.

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Author contributions

R.O.B.B. executed all experimental procedures, conception and design of study; acquisitions, analysis and of data as well as drafting the manuscript. R.R.S.Q., A.R., Y.R., M.S.B.B., and C.T.C. participated in one or more of the following: experimental procedures, and analysis and of data as well as drafting the manuscript. C.S.C. participated in funding acquisition. R.I.S. obtained the funding and participated of conception and design of study analysis and of data as well as drafting the manuscript. All authors participated in the redaction of the manuscript and revised it. All authors read and approved the final manuscript.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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